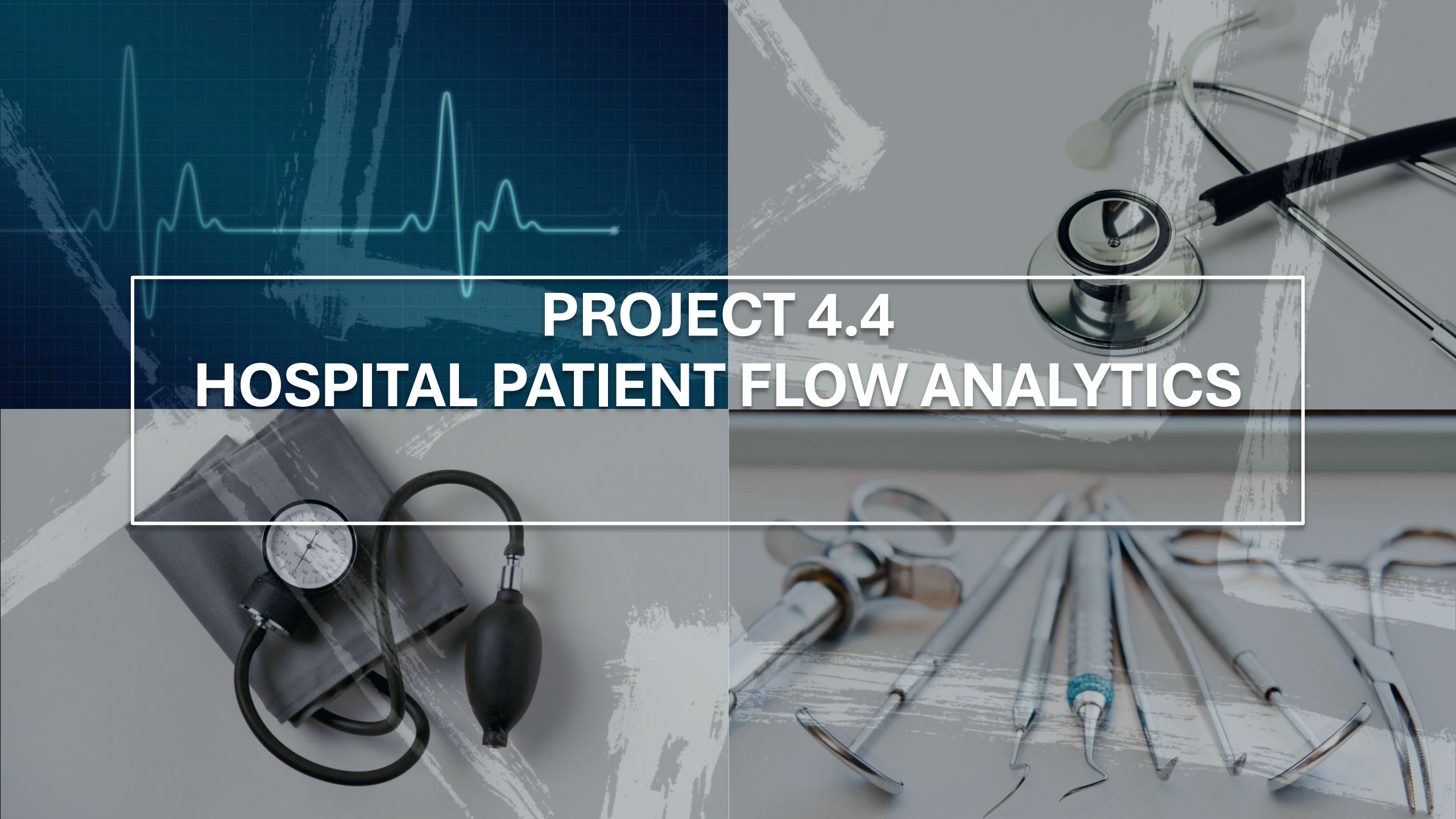


The background is a collage of medical-related images. The top left shows a blue ECG line on a dark blue grid. The top right shows a silver stethoscope and surgical instruments. The bottom left shows a blood pressure cuff and pump. The bottom right shows various surgical tools like forceps and scalpels. The text is centered in a white box with a thin black border.

PROJECT 4.4

HOSPITAL PATIENT FLOW ANALYTICS

PROJECT 4.4 — HOSPITAL PATIENT FLOW ANALYTICS

Purpose:

Prepare learners for real-world multi-table analytics before

Aviation (Project 5)

And

Cybersecurity (7) .

PROJECT 4.4 — HOSPITAL PATIENT FLOW ANALYTICS

- MetroCare NHS Trust is experiencing increasing pressure on patient flow, waiting times, and bed availability. You have been hired as a **Junior Data Analyst** to investigate the causes of delays and identify opportunities to improve patient flow.
- You will analyse **four datasets**:
- **Patients**
- **Admissions**
- **Departments**
- **StaffingLevels**
- Your goal is to produce:
- A **dashboard or pivot-table report**
- A **written executive summary** (200–300 words)
- **3 evidence-based recommendations**
- **Visuals** (charts, pivots, or Power BI pages)
- You may work in **groups for discussion**, but **all outputs must be your own**.

DATASET PACK

1. Patients Table

Purpose: Demographics + patient-level attributes.

Field	Type	Description
PatientID	INT	Unique patient identifier
Age	INT	0–95
Gender	VARCHAR	Male / Female / Other
Ethnicity	VARCHAR	Standard UK categories
Postcode	VARCHAR	Partial postcode (e.g., SL3, EN3)
LongTermCondition	VARCHAR	Yes / No

DATASET PACK

2. Admissions Table

Purpose: Tracks patient flow through the hospital.

Field	Type	Description
AdmissionID	INT	Unique admission
PatientID	INT	FK – Patients
DepartmentID	INT	FK – Departments
AdmissionDate	DATE	Date admitted
DischargeDate	DATE	Date discharged
LengthOfStay	INT	Auto-calculated
AdmissionType	VARCHAR	Emergency / Elective
WaitingTimeMinutes	INT	Time before being seen
Outcome	VARCHAR	Discharged / Transferred / Deceased

DATASET PACK

3. Departments Table

Purpose: Hospital structure.

Field	Type	Description
DepartmentID	INT	Unique department
DepartmentName	VARCHAR	e.g., A&E, Cardiology
Ward	VARCHAR	Ward name
BedsAvailable	INT	Number of beds
Region	VARCHAR	Hospital region (North, South, etc.)

DATASET PACK

4. Staffing Levels Table

Purpose: Links staffing to patient flow.

Field	Type	Description
StaffingID	INT	Unique record
DepartmentID	INT	FK - Departments
Date	DATE	Daily staffing record
NursesOnShift	INT	Count
DoctorsOnShift	INT	Count
AgencyStaffUsed	VARCHAR	Yes / No
StaffShortageFlag	VARCHAR	Yes / No

Task 1 — Data Preparation

- Load all four datasets into Excel.
- Check for missing values.
- Calculate **LengthOfStay** if not provided.
- Create a new column: $\text{BedPressureScore} = \text{LengthOfStay} * (\text{WaitingTimeMinutes} / 30)$

Task 2 — Pivot Table Analysis

Create at least **three** pivot tables:

1. **Average waiting time by department**
2. **Admissions by outcome (Discharged / Transferred / Deceased)**
3. **Staffing levels vs waiting times**
 - Join StaffingLevels → Departments using VLOOKUP/XLOOKUP

Task 3 — Visualisations

Create at least **three** pivot tables:

1. **Average waiting time by department**
2. **Admissions by outcome (Discharged / Transferred / Deceased)**
3. **Staffing levels vs waiting times**
 - Join StaffingLevels → Departments using VLOOKUP/XLOOKUP

Task 4 — Insights & Recommendations

Answer the following:

1. Which departments have the **longest waiting times**?
2. Does **staff shortage** correlate with delays?
3. Which patient groups (age, condition) experience the **longest stays**?
4. What are the **top 3 causes** of poor patient flow?

Task 5 — Executive Summary (200–300 words)

Write a professional summary including:

Key findings

Patterns or trends

Risks or concerns

3 actionable recommendations

This is the part they will put in their **portfolio**.